

Adam#

<https://pytorch.org/docs/stable/generated/torch.optim.Adam.html#torch.optim.>

Description:

Implements Adam algorithm. For further details regarding the algorithm we refer to Adam: A Method for Stochastic Optimization. `params` (iterable) – iterable of parameters or `named_parameters` to optimize or iterable of dicts defining parameter groups. When using `named_parameters`, all parameters in all groups should be named `lr` (float, Tensor, optional) – learning rate (default: 1e-3). A tensor LR is not yet supported for all our implementations. Please use a float LR if you are not also specifying `fused=True` or `capturable=True`. `betas` (Tuple[float, float], optional) – coefficients used for computing running averages of gradient and its square (default: (0.9, 0.999))

Function Signature

```
class torch.optim.Adam(params, lr=0.001, betas=(0.9, 0.999),  
eps=1e-08, weight_decay=0, amsgrad=False, *, foreach=None,  
maximize=False, capturable=False, differentiable=False,  
fused=None, decoupled_weight_decay=False)[source]#
```

Parameters

```
add_param_group(param_group)[source]#
```

Parameters

```
load_state_dict(state_dict)[source]#
```

Returns:

dict[str, Any]

Code Examples

```

>>> model = torch.nn.Linear(10, 10)
>>> optim = torch.optim.SGD(model.parameters(), lr=3e-4)
>>> scheduler1 = torch.optim.lr_scheduler.LinearLR(
...     optim,
...     start_factor=0.1,
...     end_factor=1,
...     total_iters=20,
... )
>>> scheduler2 = torch.optim.lr_scheduler.CosineAnnealingLR(
...     optim,
...     T_max=80,
...     eta_min=3e-5,
... )
>>> lr = torch.optim.lr_scheduler.SequentialLR(
...     optim,
...     schedulers=[scheduler1, scheduler2],
...     milestones=[20],
... )
>>> lr.load_state_dict(torch.load("./save_seq.pt"))
>>> # now load the optimizer checkpoint after loading the
LRScheduler
>>> optim.load_state_dict(torch.load("./save_optim.pt"))

```

```
hook(optimizer) -> None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer, state_dict) -> state_dict or None
```

```
hook(optimizer) -> None
```

```
hook(optimizer, args, kwargs) -> None
```

```
hook(optimizer, args, kwargs) -> None or modified args and  
kwargs
```

```

{
  'state': {
    0: {'momentum_buffer': tensor(...), ...},
    1: {'momentum_buffer': tensor(...), ...},
    2: {'momentum_buffer': tensor(...), ...},
    3: {'momentum_buffer': tensor(...), ...}
  },
  'param_groups': [
    {
      'lr': 0.01,
      'weight_decay': 0,
      ...
      'params': [0]
      'param_names' ['param0'] (optional)
    },
    {
      'lr': 0.001,
      'weight_decay': 0.5,
      ...
      'params': [1, 2, 3]
      'param_names': ['param1', 'layer.weight',
'layer.bias'] (optional)
    }
  ]
}

```

Notes & Warnings

NOTE

Note The foreach and fused implementations are typically faster than the for-loop, single-tensor implementation, with fused being theoretically fastest with both vertical and horizontal fusion. As such, if the user has not specified either flag (i.e., when `foreach = fused = None`), we will attempt defaulting to the foreach implementation when the tensors are all on CUDA. Why not fused? Since the fused implementation is relatively new, we want to give it sufficient bake-in time. To specify fused, pass `True` for fused. To force running the for-loop implementation, pass `False` for either `foreach` or `fused`.

NOTE

Note A prototype implementation of Adam and AdamW for MPS supports `torch.float32` and `torch.float16`.

⚠ WARNING

Warning Make sure this method is called after initializing `torch.optim.lr_scheduler.LRScheduler`, as calling it beforehand will overwrite the loaded learning rates.

NOTE

Note The names of the parameters (if they exist under the “param_names” key of each param group in `state_dict()`) will not affect the loading process. To use the parameters’ names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom `register_load_state_dict_pre_hook` should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer `param_names` will remain unchanged.

Detailed Documentation

Documentation

Implements Adam algorithm.

For further details regarding the algorithm we refer to Adam: A Method for Stochastic Optimization.

`params` (iterable) – iterable of parameters or `named_parameters` to optimize or iterable of dicts defining parameter groups. When using `named_parameters`, all parameters in all groups should be named

`lr` (float, Tensor, optional) – learning rate (default: 1e-3). A tensor LR is not yet supported for all our implementations. Please use a float LR if you are not also specifying `fused=True` or `capturable=True`.

`betas` (Tuple[float, float], optional) – coefficients used for computing running averages of

gradient and its square (default: (0.9, 0.999))

eps (float, optional) – term added to the denominator to improve numerical stability (default: 1e-8)

weight_decay (float, optional) – weight decay (L2 penalty) (default: 0)

decoupled_weight_decay (bool, optional) – if True, this optimizer is equivalent to AdamW and the algorithm will not accumulate weight decay in the momentum nor variance. (default: False)

amsgrad (bool, optional) – whether to use the AMSGrad variant of this algorithm from the paper On the Convergence of Adam and Beyond (default: False)

foreach (bool, optional) – whether foreach implementation of optimizer is used. If unspecified by the user (so foreach is None), we will try to use foreach over the for-loop implementation on CUDA, since it is usually significantly more performant. Note that the foreach implementation uses $\sim \text{sizeof}(\text{params})$ more peak memory than the for-loop version due to the intermediates being a tensorlist vs just one tensor. If memory is prohibitive, batch fewer parameters through the optimizer at a time or switch this flag to False (default: None)

maximize (bool, optional) – maximize the objective with respect to the params, instead of minimizing (default: False)

capturable (bool, optional) – whether this instance is safe to capture in a graph, whether for CUDA graphs or for torch.compile support. Tensors are only capturable when on supported accelerators. Passing True can impair ungraphed performance, so if you don't intend to graph capture this instance, leave it False (default: False)

differentiable (bool, optional) – whether autograd should occur through the optimizer step in training. Otherwise, the step() function runs in a torch.no_grad() context. Setting

to True can impair performance, so leave it False if you don't intend to run autograd through this instance (default: False)

fused (bool, optional) – whether the fused implementation is used. Currently, torch.float64, torch.float32, torch.float16, and torch.bfloat16 are supported. (default: None)

The foreach and fused implementations are typically faster than the for-loop, single-tensor implementation, with fused being theoretically fastest with both vertical and horizontal fusion. As such, if the user has not specified either flag (i.e., when foreach = fused = None), we will attempt defaulting to the foreach implementation when the tensors are all on CUDA. Why not fused? Since the fused implementation is relatively new, we want to give it sufficient bake-in time. To specify fused, pass True for fused. To force running the for-loop implementation, pass False for either foreach or fused.

A prototype implementation of Adam and AdamW for MPS supports torch.float32 and torch.float16.

Add a param group to the Optimizer's param_groups.

This can be useful when fine tuning a pre-trained network as frozen layers can be made trainable and added to the Optimizer as training progresses.

param_group (dict) – Specifies what Tensors should be optimized along with group specific optimization options.

Load the optimizer state.

state_dict (dict) – optimizer state. Should be an object returned from a call to state_dict().

Make sure this method is called after initializing torch.optim.lr_scheduler.LRScheduler,

as calling it beforehand will overwrite the loaded learning rates.

The names of the parameters (if they exist under the “param_names” key of each param group in `state_dict()`) will not affect the loading process. To use the parameters’ names for custom cases (such as when the parameters in the loaded state dict differ from those initialized in the optimizer), a custom `register_load_state_dict_pre_hook` should be implemented to adapt the loaded dict accordingly. If `param_names` exist in loaded state dict `param_groups` they will be saved and override the current names, if present, in the optimizer state. If they do not exist in loaded state dict, the optimizer `param_names` will remain unchanged.

Register a `load_state_dict` post-hook which will be called after `load_state_dict()` is called. It should have the following signature:

The `optimizer` argument is the optimizer instance being used.

The hook will be called with argument `self` after calling `load_state_dict` on `self`. The registered hook can be used to perform post-processing after `load_state_dict` has loaded the `state_dict`.

`hook (Callable)` – The user defined hook to be registered.

`prepend (bool)` – If `True`, the provided post hook will be fired before all the already registered post-hooks on `load_state_dict`. Otherwise, the provided hook will be fired after all the already registered post-hooks. (default: `False`)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a `load_state_dict` pre-hook which will be called before `load_state_dict()` is called. It should have the following signature:

The `optimizer` argument is the optimizer instance being used and the `state_dict` argument is a shallow copy of the `state_dict` the user passed in to `load_state_dict`. The hook may modify the `state_dict` inplace or optionally return a new one. If a `state_dict` is returned, it will be used to be loaded into the optimizer.

The hook will be called with argument `self` and `state_dict` before calling `load_state_dict` on `self`. The registered hook can be used to perform pre-processing before the `load_state_dict` call is made.

`hook (Callable)` – The user defined hook to be registered.

`prepend (bool)` – If `True`, the provided pre hook will be fired before all the already registered pre-hooks on `load_state_dict`. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: `False`)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a state dict post-hook which will be called after `state_dict()` is called.

It should have the following signature:

The hook will be called with arguments `self` and `state_dict` after generating a `state_dict` on `self`. The hook may modify the `state_dict` inplace or optionally return a new one. The registered hook can be used to perform post-processing on the `state_dict` before it is returned.

`hook (Callable)` – The user defined hook to be registered.

`prepend (bool)` – If `True`, the provided post hook will be fired before all the already registered post-hooks on `state_dict`. Otherwise, the provided hook will be fired after all

the already registered post-hooks. (default: False)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register a state dict pre-hook which will be called before `state_dict()` is called.

It should have the following signature:

The optimizer argument is the optimizer instance being used. The hook will be called with argument `self` before calling `state_dict` on `self`. The registered hook can be used to perform pre-processing before the `state_dict` call is made.

hook (Callable) – The user defined hook to be registered.

prepend (bool) – If True, the provided pre hook will be fired before all the already registered pre-hooks on `state_dict`. Otherwise, the provided hook will be fired after all the already registered pre-hooks. (default: False)

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemoveableHandle`

Register an optimizer step post hook which will be called after optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemovableHandle`

Register an optimizer step pre hook which will be called before optimizer step.

It should have the following signature:

The optimizer argument is the optimizer instance being used. If args and kwargs are modified by the pre-hook, then the transformed values are returned as a tuple containing the new_args and new_kwargs.

hook (Callable) – The user defined hook to be registered.

a handle that can be used to remove the added hook by calling `handle.remove()`

`torch.utils.hooks.RemovableHandle`

Return the state of the optimizer as a dict.

It contains two entries:

differs between optimizer classes, but some common characteristics hold. For example, state is saved per parameter, and the parameter itself is NOT saved. state is a Dictionary mapping parameter ids to a Dict with state corresponding to each parameter.

parameter group is a Dict. Each parameter group contains metadata specific to the optimizer, such as learning rate and weight decay, as well as a List of parameter IDs of the parameters in the group. If a param group was initialized with `named_parameters()` the names content will also be saved in the state dict.

NOTE: The parameter IDs may look like indices but they are just IDs associating state with param_group. When loading from a state_dict, the optimizer will zip the param_group params (int IDs) and the optimizer param_groups (actual `nn.Parameter s`)

in order to match state WITHOUT additional verification.

A returned state dict might look something like:

Perform a single optimization step.

closure (Callable, optional) – A closure that reevaluates the model and returns the loss.

Reset the gradients of all optimized torch.Tensor s.

set_to_none (bool, optional) –

Instead of setting to zero, set the grads to None. Default: True

This will in general have lower memory footprint, and can modestly improve performance. However, it changes certain behaviors. For example:

When the user tries to access a gradient and perform manual ops on it, a None attribute or a Tensor full of 0s will behave differently.

If the user requests zero_grad(set_to_none=True) followed by a backward pass, .grads are guaranteed to be None for params that did not receive a gradient.

torch.optim optimizers have a different behavior if the gradient is 0 or None (in one case it does the step with a gradient of 0 and in the other it skips the step altogether).